Conditional Stationarity and Positive Curvature of the Spectral Trace at the Critical Line

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Abstract

Building on the curvature identity $O''(\frac{1}{2}) = -2B$ established in [1] and the trace formula $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ proved in [2], we address two of the five open problems of Paper 6 concerning the first derivative $O'(\frac{1}{2})$ and the pointwise asymptotic behaviour of the Guinand–Weil constant term. Under an explicit subleading archimedean hypothesis (Assumption 3.2, numerically supported) and a proxy-transfer hypothesis (Assumption 3.3, formally weaker in content than RH: RH implies it, no converse claimed), we prove that the ratio $r_p^\infty := \lim_{\varepsilon\to 0^+} |\mathrm{Const}_p^\infty(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ equals $\frac{1}{2}$ for every prime p, and derive an exact three-term bookkeeping decomposition of $O'(\frac{1}{2})$ with two explicit leading terms and a defined residual. We further prove under the same assumptions that for every prime cutoff $\kappa \geq 2$ there exists $\varepsilon_{\mathrm{int}}(\kappa) > 0$ such that the integrated bias $B_{\mathrm{int}}^\infty(\kappa, \varepsilon) < 0$ for all $\varepsilon \in (0, \varepsilon_{\mathrm{int}}(\kappa))$. Under an additional logarithmic-saving hypothesis on prime exponential sums (Assumption (A^{**}), Cancellation Hypothesis) we establish a conditional constant-factor bound on $|O'(\frac{1}{2})|$ relative to the trivial bound (Proposition 7.1). The full conditional local-selection statement (strict local minimum of O at $\sigma = \frac{1}{2}$) additionally requires the Weighted Bias Bridge hypothesis $B(\kappa, \varepsilon, N) < 0$ and exact stationarity $O'(\frac{1}{2}) = 0$ (Corollary 7.3). No hypothetical input is used in the unconditional results: no Riemann Hypothesis, no Hilbert–Pólya postulate, no GUE conjecture.

What is established unconditionally. Definition 4.1 (bookkeeping decomposition); Proposition 8.1 (antisymmetry). Observation 4.2 provides numerical values at reference parameters and is reported under *What is numerical.*

What is proved under Assumptions 3.2 and 3.3. For every prime p, $r_p^\infty = \frac{1}{2}$ (Theorem 3.1); in particular $Z_{p,\infty}^+(\varepsilon) < 0$ for all sufficiently small $\varepsilon > 0$ (Corollary 3.7); and for every $\kappa \geq 2$ there exists $\varepsilon_{\mathrm{int}}(\kappa) > 0$ such that $B_{\mathrm{int}}^\infty(\kappa, \varepsilon) < 0$ for $\varepsilon \in (0, \varepsilon_{\mathrm{int}}(\kappa))$ (Proposition 6.1).

What is numerical. At $(\kappa, \varepsilon, N) = (53, 0.05, 100)$: $O'(\frac{1}{2}) = +2.4751$; $O''(\frac{1}{2}) = +38\,685.1$; $B = -19\,342.5$. The trivial bound $\mathcal{W}_{\varepsilon,N} \cdot P(\kappa) \approx 1277$ exceeds $|O'(\frac{1}{2})|$ by approximately $516\times$ $(|O'(\frac{1}{2})|/(\mathcal{W}_{\varepsilon,N} \cdot P(\kappa)) \approx 0.00194)$. Separately, the internal three-term balance satisfies $|T_{\mathrm{pol}} + T_{\mathrm{main}}|/|O'(\frac{1}{2})| \approx 15.7$ (see Observation 4.2 for the explicit values). $\varepsilon_{\mathrm{int}}(53) > 0.200$; $O''(\frac{1}{2}) = +38\,685.1 > 0$ [numerical].

Additional numerical diagnostics. Sections 5, 8, and 9 report further finite-grid numerical diagnostics, including the value $O'(\frac{1}{2}; 101, 0.05, 100) = -137.6$ used in the structural-necessity discussion, observed sign-region data for $\varepsilon_{\mathrm{int}}(\kappa)$, and the tested near-minimum behaviour at $\kappa = 53$ and $\kappa = 101$ (fixed $N = 100$). These are finite-grid numerical observations and are not used as proved inputs. The strong κ-dependence of $O'(\frac{1}{2}; \kappa, \varepsilon, N)$ between $\kappa = 53$ and $\kappa = 101$ is precisely the empirical signature that motivates introducing the

cancellation hypothesis (A^{**}): exact stationarity at the reference parameters is not claimed to be universal, and the variation across κ is the structural reason why an analytical input on prime-exponential-sum cancellation is required to control $|O'(\frac{1}{2})|$ uniformly.

What is conditional. Under Assumption (A^{**}) (Cancellation Hypothesis, § 5): a constant-factor bound on $|O'(\frac{1}{2}; \kappa, \varepsilon, N)|$ relative to the trivial bound (Proposition 7.1). Under the additional hypotheses $B(\kappa, \varepsilon, N) < 0$ and $O'(\frac{1}{2}) = 0$ (exact stationarity), this gives a conditional local-selection criterion at $\sigma = \frac{1}{2}$ (Corollary 7.3).

What remains open. The Weighted Bias Bridge — whether $B^{\infty}_{\text{int}} < 0$ implies $O''(\frac{1}{2}) > 0$ via $O''(\frac{1}{2}) = -2B$ (Open Problem 9.1); the Cancellation Hypothesis (A^{**}) (Open Problem 9.2); the Guinand–Weil bridge with off-critical control (Open Problem 9.3); the scaling of $\varepsilon_{\text{int}}(\kappa)$ (Open Problem 9.4); the near-minimum question (Open Problem 9.5); and the Weil-transfer connection (Open Problem 9.6).

Structure. The unconditional part (Definition 4.1, Proposition 8.1) is proof-complete and non-circular. Results conditional on Assumptions 3.2 and 3.3 are clearly labelled. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made in §§ 2–6; Assumption (A^{**}) in § 7 is explicitly declared conditional.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The trace parameter $\sigma \in \mathbb{R}$ is evaluated at $\sigma = \frac{1}{2}$ unless stated otherwise.

Global parameters. The reference values are $\kappa = 53$ (prime cutoff, yielding 16 active primes $p \leq \kappa$), $\varepsilon = 0.05$ (Gaussian regularisation parameter), and $N = 100$ (number of Riemann ordinates γ_k employed). These values are fixed throughout unless explicitly varied. The key weight sums are

$$\mathcal{W}_{\varepsilon,N} := \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\,\gamma_k, \qquad A(\varepsilon, N) := \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2},$$
$$P(\kappa) := \sum_{p\leq\kappa} \log p \qquad \text{(the Chebyshev theta sum).}$$

At the reference parameters, $\mathcal{W}_{0.05,100} \approx 28.42$ and $P(53) \approx 44.93$.

Main objects. The coupling operator $\Phi(\sigma) : H_{\text{str}} \to H_{\text{null}}$ is defined by

$$\Phi(\sigma)_{k,p} = e^{-\varepsilon^2\gamma_k^2/2} \sin(\sigma\gamma_k \log p),$$

and the Tehrani operator is $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$. The trace oscillatory sum is

$$O(\sigma) = \tfrac{1}{2} \sum_{k,p} e^{-\varepsilon^2\gamma_k^2} \cos(2\sigma\gamma_k \log p),$$

with derivatives $O'(\sigma)$ and $O''(\sigma)$ taken with respect to σ.

Curvature and bias. The direct curvature sum is

$$B := \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p),$$

and the integrated bias is

$$B^+_{\mathrm{int}}(\varepsilon, N) := \sum_{p\le\kappa} (\log p)^2 Z^+_{p,N}(\varepsilon).$$

Ordinate proxies. The finite proxy used in numerics is

$$Z^+_{p,N}(\varepsilon) := \operatorname{Re} Z_p(\varepsilon, N), \qquad Z_p(\varepsilon, N) := \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2} p^{i\gamma_k}.$$

The asymptotic analogue is

$$Z^+_{p,\infty}(\varepsilon) := \sum_{k\ge1} e^{-\varepsilon^2\gamma_k^2} \cos(\gamma_k \log p),$$

and the full signed asymptotic sum is $Z_p^\infty(\varepsilon) := 2Z^+_{p,\infty}(\varepsilon)$. The full Guinand–Weil object, summed over all non-trivial zeros, is

$$\widetilde{Z}_p^{\mathrm{GW,full}}(\varepsilon) := \sum_\rho h^{\mathrm{even}}_{p,\varepsilon}((\rho - \tfrac{1}{2})/i),$$

where $h^{\mathrm{even}}_{p,\varepsilon}(u) = e^{-\varepsilon^2u^2}\cos(u \log p)$.

Leading term and exponential sum. The Guinand–Weil leading term is

$$\mathrm{Main}_p(\varepsilon) := -\frac{\log p}{2\sqrt{\pi}\,\varepsilon\sqrt{p}}.$$

The constant term Const_p^∞ used in Theorem 3.1 is defined locally there. The exponential sum over primes is

$$S_k(\kappa) := \sum_{p\le\kappa} (\log p)\sin(\gamma_k \log p).$$

Remark. *The symbol $S_k(\kappa)$ is distinct from $M_k(\kappa)$ of Paper 4 [8], which denotes the partial-sum maximum $\max_{y\le\kappa} |\sum_{p\le y} \sin(\gamma_k \log p)|$. Both symbols appear in the series; the present paper uses $S_k(\kappa)$ to avoid ambiguity.*

Analytical foundations. The proof of Theorem 3.1 uses the exact Guinand–Weil decomposition of [1] and classical Gaussian Fourier analysis. Gaussian decay together with standard Riemann–von Mangoldt zero-counting estimates [18, Thm. 5.8] justifies convergence of the infinite ordinate proxies and the $N \to \infty$ form of the derivative bound; these estimates are not used as sign input in the asymptotic proof of Theorem 3.1. Lemma 3.6 is a separate prime-power Gaussian tail estimate that does not invoke zero-density. The conditional selection criterion (Proposition 7.1, Corollaries 7.2 and 7.3) uses the trace formula of [2] and the curvature identity of [1].

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of working precision; Riemann zeros γ_k were evaluated via `mpmath.zetazero`. Verification code is available at [3] (`https://github.com/utehrani/analysislab-nt`).

Critical distinctions. The following symbols look similar but are distinct objects (referenced throughout the paper):

NOT	**BUT**
$Z^+_{p,N}$ (finite, numerics)	$Z^+_{p,\infty}$ (infinite, Thm 3.1)
$Z^\infty_p = 2Z^+_{p,\infty}$ (proxy, factor 2)	$\widetilde{Z}^{\mathrm{GW,full}}_p$ (full GW, all zeros)
B (γ_k^2-weighted, $-19\,342.5$)	B^+_{int} (unweighted, -42.21)
$\mathcal{W}_{\varepsilon,N}$ (γ_k-weighted, this paper)	$A(\varepsilon, N)$ (unweighted, Paper 5)
Const_p (finite-N, Papers 5/6)	Const^∞_p (this paper, Thm 3.1)
$r_p(\varepsilon) = 0.75$–0.80 at $\varepsilon = 0.05$	$r^\infty_p = \frac{1}{2}$ ($\varepsilon \to 0^+$ limit; Thm 3.1)
$D_{\mathrm{SEL}} \approx 10.985$ (Paper 5)	$D \approx 9.470$ (Paper 2)

Truncation: $|Z^+_{p,\infty} - Z^+_{p,N}|/|\mathrm{Main}_p| \lesssim 10^{-61}$ at reference parameters (truncation bound, Paper 6, § 3). The identity $\widetilde{Z}^{\mathrm{GW,full}}_p = Z^\infty_p$ holds when all non-trivial zeros lie on the critical line (Paper 5, § 6; Assumption 3.3).

1 Introduction

1.1 Background and Motivation

This paper is the seventh in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function. The central construction is a finite-dimensional Hilbert-space model in which prime numbers mediate coupling between zeros, encoded in an explicit family of operators whose spectral properties reflect arithmetic structure at the critical line.

Paper 1 [4] introduced the local curvature function $H_{\text{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma)$, where $V_p(\sigma)$ is the prime-p contribution to the spectral curvature. At $\sigma = \frac{1}{2}$ the sum diverges like $2(\log \kappa)^2$ as $\kappa \to \infty$, identifying the critical line as a phase boundary through a structural divergence of the curvature.

Paper 2 [5] connected this local curvature to the Weil explicit functional, constructing σ-dependent admissible Gaussian test functions $g^*_{\sigma,\varepsilon}$ for which the curvature divergence translates into the growth of the diagonal energy partial sums. In the renormalised critical-line scale, $\sum_{p \leq \kappa} f_p$ converges to the diagonal energy $D := \sum_p f_p \approx 9.470$. The Weil positivity framework of [6] was identified as the natural target for future operator-theoretic results.

Paper 3 [7] made the geometric structure explicit by introducing two finite-dimensional Hilbert spaces, H_{str} over primes and H_{null} over zeros, together with the coupling operator $\Phi : H_{\text{str}} \to H_{\text{null}}$ defined by the prime-zero phasor expression $\Phi_{k,p} = e^{-\varepsilon^2 \gamma_k^2/2} \sin(\gamma_k \log p)$. The later one-parameter convention $\sin(\sigma \gamma_k \log p)$ is introduced in Paper 5 and is the convention used throughout the present paper. The loop operator $T = \Phi^* \circ \Phi$ on H_{str} and the energy asymmetry ratio $\eta_{\text{orig}} > 0$ were introduced, with the latter confirmed numerically.

Paper 4 [8] introduced the dual operator $\widetilde{T} = \Phi \circ \Phi^*$ on H_{null} — the Tehrani operator — and proved conditional energy asymmetry under an explicit remainder-control assumption. A Hilbert–Pólya diagnostic yielded $r_2 \to 0.16$ as $\kappa \to \infty$ [numerical], numerically falsifying a Hilbert–Pólya interpretation of $\widetilde{T}$ on the tested grid and distinguishing it from the constructions of Berry and Keating [9] and de Branges [10].

Paper 5 [2] extended the operator family to a one-parameter family $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ and proved the exact spectral trace identity

$$\operatorname{Tr}(\widetilde{T}(\sigma)) \;=\; D_{\text{SEL}} - O(\sigma), \qquad O(\sigma) \;=\; \tfrac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p),$$

where D_{SEL} is a σ-independent constant. The paper introduced the Guinand–Weil smoothed zero sum $\widetilde{Z}_p^{\text{GW,full}}$ as formal motivation for analysing the bias of prime–zero sums, and formulated the Bias Conjecture for the infinite ordinate proxy: $Z_p^{\infty}(\varepsilon) < 0$ for all primes $p \leq \kappa$ at small enough ε. The transfer from the full Guinand–Weil object to the ordinate proxy was left open as the GW bridge.

Paper 6 [1] proved the curvature–bias identity $O''(\frac{1}{2}) = -2B$ by direct differentiation of the trace formula of Paper 5, where $B = \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p)$. The leading term $\mathrm{Main}_p(\varepsilon)$ of the Guinand–Weil decomposition was proved strictly negative for every prime p and every $\varepsilon > 0$, and the numerical value $B = -19\,342.5$ at the reference parameters gives $O''(\frac{1}{2}) = +38\,685.1 > 0$. Five open problems were formulated, of which the present paper addresses the first two open problems of Paper 6: conditional stationarity of $O'(\frac{1}{2})$ and the pointwise asymptotic bias of $Z^{+}_{p,\infty}$.

What is new in the present paper. Papers 5–6 *conjectured* that $Z_p^{\infty}(\varepsilon) < 0$ in the limit $\varepsilon \to 0^+$ and tested it numerically. The present paper *proves the exact asymptotic rate* $r_p^{\infty} = \frac{1}{2}$ as a pointwise identity in p, conditional on two named hypotheses (Assumptions 3.2 and 3.3), with an explicit decomposition of $\mathrm{Const}_p^{\infty}$ into a pole term, the dominant $-\frac{1}{2}\mathrm{Main}_p$, an exponentially small reflected peak, and a residual that is $o(|\mathrm{Main}_p|)$. The contribution is the precise classification of orders of magnitude in this identity — not a new arithmetic conjecture.

The operator $\widetilde{T}(\sigma)$ constructed in this series is a Gram-type operator with explicit arithmetic matrix entries; it is not a Hilbert–Pólya operator (Paper 4), and does not arise from an adelic or non-commutative geometric construction. Connes and Consani [11] and Connes, Consani, and Moscovici [12] have constructed spectral triples and zeta-cycles on adelic spaces whose spectra encode the zeros of ζ via abstract operator-algebraic data. Our framework is finite-cutoff, explicit, and operates within classical Hilbert space theory over $\mathbb{C}^P$ and $\mathbb{C}^N$; the bridge to the Weil positivity criterion of Lagarias [6] and Bombieri [13] constitutes our principal open problem (Open Problem 9.6). Related approaches via the Hilbert space structure of the Weil distribution have been pursued by Burnol [14], Suzuki [15], and others.

The present paper addresses the analytic question at the core of Paper 6: whether a structural bound on $|O'(\frac{1}{2})|$ relative to $\mathcal{W}_{\varepsilon,N} \cdot P(\kappa)$ can be derived from a structural property of the prime-zero interaction. Under the stated transfer and subleading hypotheses, we obtain an asymptotic answer: the ratio $r_p^{\infty} = \frac{1}{2}$ is universal (Theorem 3.1), and the integrated-bias window is conditional on a subleading archimedean estimate (Assumption 3.2) and a proxy-transfer condition (Assumption 3.3), via Proposition 6.1. The stationarity bound requires a logarithmic-saving hypothesis on exponential sums, which we formulate precisely as Assumption (A^{**}). We have not obtained an unconditional bound of the form required by Proposition 7.1 from the five standard routes discussed in § 5. These are methodological observations on the present formulation, not impossibility results. Assumption (A^{**}) is therefore taken as the natural conditional input (§ 5, Remark 5).

1.2 Main Results

The four main results of this paper are as follows.

The first, in § 3 (Theorem 3.1), is the *pointwise asymptotic limit*: for every prime $p \geq 2$, $r_p^{\infty} := \lim_{\varepsilon \to 0^+} |\mathrm{Const}_p^{\infty}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)| = \frac{1}{2}$. This is conditional on Assumptions 3.2 and 3.3, and resolves the pointwise asymptotic bias question of Paper 6, § 8, for the ordinate proxy; the GW bridge to the full object $\widetilde{Z}_p^{\mathrm{GW,full}}$ remains open (Open Problem 9.3).

The second, in § 4 (Definition 4.1), is the *three-term decomposition* of $O'(\frac{1}{2})$: the algebraic definition $O'(\frac{1}{2}) = T_{\text{pol}}(\varepsilon) + T_{\text{main}}(\varepsilon) + T_{\text{res}}(\kappa, \varepsilon, N)$ where $T_{\text{pol}}(\varepsilon)$ and $T_{\text{main}}(\varepsilon)$ arise from the GW pole and prime main-term contributions and are algebraically explicit, while T_{res} is the residual (Definition 4.1). At the reference parameters, $|O'(\frac{1}{2})| = 2.4751$ is only approximately 0.2% of the trivial bound $\mathcal{W}_{\varepsilon,N} \cdot P(\kappa) \approx 1277$, with $T_{\text{pol}} + T_{\text{main}} = +38.84$ and $T_{\text{res}} = -36.37$ cancelling to a factor of 15.7 above $|O'(\frac{1}{2})|$ [numerical].

The third, in § 6 (Proposition 6.1), is the *integrated-bias window*: for every prime cutoff $\kappa \geq 2$ there exists $\varepsilon_{\text{int}}(\kappa) > 0$ such that $B^{\infty}_{\text{int}}(\kappa, \varepsilon) < 0$ for all $\varepsilon \in (0, \varepsilon_{\text{int}}(\kappa))$. This result is conditional on Assumptions 3.2 and 3.3, and addresses the curvature-uniformity question of Paper 6, § 8. At reference parameters $O''(\frac{1}{2}) = +38\,685.1 > 0$ [numerical].

The fourth, in § 7 (Proposition 7.1 and Corollaries 7.2, 7.3), is the *conditional selection criterion*: under Assumption (A^{**}) (Cancellation Hypothesis, § 5), $|O'(\frac{1}{2}; \kappa, \varepsilon, N)| \leq C \cdot \mathcal{W}_{\varepsilon,N} \cdot P(\kappa)$ where $C := C_0/(\log \gamma_1)^A$ and $\log \gamma_1 \approx 2.6486$ is independent of κ, ε, and N. For $\varepsilon \in (0, \varepsilon_{\text{int}}(\kappa))$ this gives a conditional derivative bound at $\sigma = \frac{1}{2}$ (Proposition 7.1); genuine smallness requires the additional condition $C_0/(\log \gamma_1)^A < 1$. We make no claim that this condition is automatically met: verifying or proving $C_0/(\log \gamma_1)^A < 1$ for an explicit (C_0, A) is itself part of the analytical content of Assumption (A^{**}). The present paper isolates this requirement; resolving it is left open (see § 9). Combined with $B^{\infty}_{\text{int}} < 0$ (Corollary 7.2) and the additional hypotheses $B(\kappa, \varepsilon, N) < 0$ and exact stationarity $O'(\frac{1}{2}) = 0$, this yields a conditional local-selection criterion at $\sigma = \frac{1}{2}$ (Corollary 7.3).

1.3 Reader Contract

This paper *establishes unconditionally*: Definition 4.1 (bookkeeping decomposition); Proposition 8.1 (antisymmetry). Observation 4.2 (numerical values at reference parameters) is numerical, not a proved statement.

This paper *proves under Assumptions 3.2 (numerically supported, analytically open) and 3.3 (proxy-transfer, formally weaker in content than RH: RH implies it, no converse claimed)*: Theorem 3.1 ($r_p^{\infty} = \frac{1}{2}$ for every prime p); Corollary 3.7 ($Z^{+}_{p,\infty}(\varepsilon) < 0$ for small ε); Proposition 6.1 (integrated-bias window $\varepsilon_{\text{int}}(\kappa) > 0$ for all $\kappa \geq 2$).

Under Assumption (A^{**}) (§ 7, Proposition 7.1): $|O'(\frac{1}{2})| \leq C \cdot \mathcal{W}_{\varepsilon,N} \cdot P(\kappa)$. Under the additional hypotheses $B(\kappa, \varepsilon, N) < 0$ and $O'(\frac{1}{2}) = 0$ (Corollary 7.3): $\sigma = \frac{1}{2}$ is a strict local minimum of O.

This paper *establishes numerically* at $(\kappa, \varepsilon, N) = (53, 0.05, 100)$: $O'(\frac{1}{2}) = +2.4751$; $O''(\frac{1}{2}) = +38\,685.1$; $B = -19\,342.5$; $\varepsilon_{\text{int}}(53) > 0.200$; the trivial bound exceeds $|O'(\frac{1}{2})|$ by approximately $516\times$, while the internal three-term balance gives $|T_{\text{pol}} + T_{\text{main}}|/|O'(\frac{1}{2})| \approx 15.7$.

This paper *leaves open*: the Weighted Bias Bridge (Open Problem 9.1); Assumption (A^{**}) (Open Problem 9.2); the Guinand–Weil bridge with off-critical control (Open Problem 9.3); the scaling of $\varepsilon_{\text{int}}(\kappa)$ (Open Problem 9.4); the near-minimum question (Open Problem 9.5); and the Weil-transfer connection (Open Problem 9.6).

2 Background: Trace Formula and Guinand–Weil Framework

2.1 The Operator Family and Trace Formula

The two-parameter coupling operator $\Phi(\sigma) : H_{\mathrm{str}} \to H_{\mathrm{null}}$ is defined on the finite-dimensional spaces $H_{\mathrm{str}} \cong \mathbb{C}^P$ (over primes $p \le \kappa$) and $H_{\mathrm{null}} \cong \mathbb{C}^N$ (over zeros $\gamma_1, \ldots, \gamma_N$) by

$$\Phi(\sigma)_{k,p} \;=\; e^{-\varepsilon^2\gamma_k^2/2}\,\sin(\sigma\gamma_k \log p).$$

The Tehrani operator is $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$, a self-adjoint endomorphism of H_{null}. Paper 5 [2] proves the following exact identity.

Theorem 2.1 (Trace formula; proved in [2])**.** *For all $\sigma \in \mathbb{R}$,*

$$\mathrm{Tr}(\widetilde{T}(\sigma)) \;=\; D_{\mathrm{SEL}} - O(\sigma),$$

where $D_{\mathrm{SEL}} := \frac{1}{2}A(\varepsilon, N)\cdot\pi(\kappa)$ is σ-independent and

$$O(\sigma) := \tfrac{1}{2}\sum_{k=1}^{N}\sum_{p\le\kappa} e^{-\varepsilon^2\gamma_k^2}\cos(2\sigma\gamma_k\log p).$$

At the reference parameters $D_{\mathrm{SEL}} \approx 10.985$.

Differentiating twice in σ yields the curvature

$$O''(\sigma) = -2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k\log p)^2\cos(2\sigma\gamma_k\log p),$$

and at $\sigma = \frac{1}{2}$ this equals $-2B$ by Paper 6 [1].

2.2 The Curvature Identity

Theorem 2.2 (Curvature–bias identity; proved in [1])**.** *The direct curvature sum $B := \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k\log p)^2\cos(\gamma_k\log p)$ satisfies the algebraic identity*

$$O''\left(\tfrac{1}{2}\right) \;=\; -2B.$$

At the reference parameters $B = -19\,342.5$ and $O''(\frac{1}{2}) = +38\,685.1$ [numerical].

2.3 The Guinand–Weil Framework

The Guinand–Weil explicit formula [16, 17] applied to the even test function $h^{\mathrm{even}}_{p,\varepsilon}(u) = e^{-\varepsilon^2u^2}\cos(u\log p)$ yields a decomposition of the formal sum over non-trivial zeros, following the conventions of [18]. For the ordinate proxy $Z^+_{p,N}(\varepsilon) = \mathrm{Re}\, Z_p(\varepsilon, N)$ at finite truncation level N

one obtains

$$Z^{+}_{p,N}(\varepsilon) = \underbrace{\tfrac{1}{2}\,\mathrm{POL}(\varepsilon,p) - \mathrm{PRIMS}_{n=p}(\varepsilon)}_{\text{leading}} - \mathrm{PRIMS}_{n\neq p}(\varepsilon) + \Gamma_p(\varepsilon) + \tfrac{1}{2}\,\mathrm{CST}(\varepsilon) + R_{p,N}(\varepsilon). \quad (1)$$

The leading terms are: $\frac{1}{2}\,\mathrm{POL}(\varepsilon,p) = e^{\varepsilon^2/4}\cosh(\log p/2)$ (Lemma 3.4); $-\mathrm{PRIMS}_{n=p}(\varepsilon) = \mathrm{Main}_p(\varepsilon)\cdot\frac{1}{2}(1+e^{-(\log p)^2/\varepsilon^2})$ (Lemma 3.5), where $\mathrm{Main}_p(\varepsilon) := -(\log p)/(2\sqrt{\pi}\,\varepsilon\sqrt{p})$. The off-diagonal contribution $\mathrm{PRIMS}_{n\neq p}$ is exponentially small (Lemma 3.6). The archimedean contribution Γ_p and constant term $\frac{1}{2}\,\mathrm{CST}$ are subleading under Assumption 3.2. The remainder $R_{p,N}$ is negligible at reference parameters (truncation estimate of [1]).

For Theorem 3.1 we work with the infinite analogue

$$\mathrm{Const}^{\infty}_p(\varepsilon) := Z^{+}_{p,\infty}(\varepsilon) - \mathrm{Main}_p(\varepsilon) - \Gamma_p(\varepsilon),$$

where $Z^{+}_{p,\infty}(\varepsilon) := \sum_{k\geq 1} e^{-\varepsilon^2\gamma_k^2}\cos(\gamma_k \log p)$.

Remark. *The identity $\widetilde{Z}^{\mathrm{GW,full}}_p(\varepsilon) = Z^{\infty}_p(\varepsilon)$ holds when all non-trivial zeros lie on the critical line; without this assumption an off-critical contribution must be added (Paper 5, § 6). All computations in this paper use the ordinate proxy $Z^{+}_{p,N}(\varepsilon)$.*

3 The Pointwise Asymptotic Limit

Theorem 3.1 (Pointwise asymptotic limit; proved under Assumptions *3.2* and *3.3*)**.** *For every prime $p \geq 2$,*

$$r^{\infty}_p \;:=\; \lim_{\varepsilon\to 0^+} \frac{|\mathrm{Const}^{\infty}_p(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|} \;=\; \frac{1}{2},$$

where $\mathrm{Const}^{\infty}_p(\varepsilon) := Z^{+}_{p,\infty}(\varepsilon) - \mathrm{Main}_p(\varepsilon) - \Gamma_p(\varepsilon)$. The limit is independent of p.

Assumption 3.2 (Subleading archimedean terms)**.** *For every prime $p \geq 2$, as $\varepsilon \to 0^+$,*

$$|\Gamma_p(\varepsilon)| + \tfrac{1}{2}|\mathrm{CST}(\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)\,.$$

Remark. *Assumption 3.2 is numerically supported but not proved. For the gamma contribution, Paper 6 reports finite-grid numerical evidence that the ratio $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)|$ is consistent with linear scaling in ε on the tested grid: across all 16 primes $p \leq 53$ and five values $\varepsilon \in \{0.02, 0.03, 0.05, 0.08, 0.12\}$, the log-log slope is 1.001 ± 0.003. This is evidence for $|\Gamma_p(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)| = O_{\mathrm{num}}(\varepsilon)$, not an analytic proof of a limit as $\varepsilon \to 0^+$. Since $|\mathrm{Main}_p(\varepsilon)| \asymp \varepsilon^{-1}$, ratio scaling of this form would correspond to $|\Gamma_p(\varepsilon)| = O_{\mathrm{num}}(1)$, not to $|\Gamma_p(\varepsilon)| = O(\varepsilon)$. The estimate $|\Gamma_p(\varepsilon)| + \frac{1}{2}|\mathrm{CST}(\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)$ therefore remains exactly the hypothesis of Assumption 3.2. The o statement for the constant-term contribution $|\mathrm{CST}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)| \to 0$ is analytically open; numerically $|\mathrm{CST}(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)| \sim \varepsilon$ [numerical, same dataset as Γ_p]. An analytic proof of the combined estimate remains open.*

Assumption 3.3 (Proxy transfer)**.** *For every fixed prime $p \geq 2$, the full Guinand–Weil zero* *sum admits a decomposition*

$$\widetilde{Z}_p^{\mathrm{GW,full}}(\varepsilon) = Z_p^{\infty}(\varepsilon) + E_p^{\mathrm{off}}(\varepsilon), \qquad Z_p^{\infty}(\varepsilon) := 2Z_{p,\infty}^{+}(\varepsilon),$$

where

$$E_p^{\mathrm{off}}(\varepsilon) = o(|\mathrm{Main}_p(\varepsilon)|) \qquad (\varepsilon \to 0^+).$$

Equivalently, the positive-ordinate proxy satisfies

$$Z_{p,\infty}^{+}(\varepsilon) = \tfrac{1}{2}\, \widetilde{Z}_p^{\mathrm{GW,full}}(\varepsilon) - \tfrac{1}{2}\, E_p^{\mathrm{off}}(\varepsilon).$$

This half-Guinand–Weil form is used in Theorem 3.1.

Remark. *Assumption 3.3 is implied by RH and is formally weaker in content: it requires only asymptotic smallness of the off-critical contribution, not its vanishing. No converse implication is claimed. No analytic proof is available without RH.*

The proof rests on three lemmas which identify the leading terms of $\mathrm{Const}_p^{\infty}(\varepsilon)$ as $\varepsilon \to 0$.

Lemma 3.4 (Pole term; proved)**.** *For every prime $p \geq 2$ and every $\varepsilon > 0$,*

$$\tfrac{1}{2}\,\mathrm{POL}(\varepsilon, p) \;=\; e^{\varepsilon^2/4} \cosh\!\left(\tfrac{\log p}{2}\right).$$

Proof. By definition, $h_{p,\varepsilon}^{\mathrm{even}}(i/2) = e^{-\varepsilon^2 (i/2)^2} \cos(i \log p/2) = e^{\varepsilon^2/4} \cosh(\log p/2)$. The evaluation at $u = -i/2$ gives the same value by the even symmetry of $h_{p,\varepsilon}^{\mathrm{even}}$. The pole contribution in the Guinand–Weil formula is $(h(i/2) + h(-i/2))/2 = e^{\varepsilon^2/4} \cosh(\log p/2)$. □

Lemma 3.5 (Prime-p main term; proved)**.** *For every prime $p \geq 2$ and every $\varepsilon > 0$,*

$$-\mathrm{PRIMS}_{n=p}(\varepsilon) \;=\; \tfrac{1}{2}\,\mathrm{Main}_p(\varepsilon) \cdot \big[1 + e^{-(\log p)^2/\varepsilon^2}\big],$$

where $\mathrm{PRIMS}_{n=p}(\varepsilon)$ denotes the $n = p$ term in the prime-power sum of the Guinand–Weil formula.

Proof. For the positive-ordinate proxy $Z_{p,\infty}^{+}$, the Guinand–Weil prime term at $n = p$ in the convention of [18, Ch. 5] contributes

$$-\frac{1}{2\pi}\,\frac{\Lambda(p)}{\sqrt{p}}\,\hat{h}_{p,\varepsilon}^{\mathrm{even}}\!\left(\frac{\log p}{2\pi}\right).$$

Substituting the Fourier evaluation $\hat{h}_{p,\varepsilon}^{\mathrm{even}}(\log p/(2\pi)) = \frac{\sqrt{\pi}}{2\varepsilon}\big(1 + e^{-(\log p)^2/\varepsilon^2}\big)$ and $\Lambda(p) = \log p$:

$$-\frac{1}{2\pi} \cdot \frac{\log p}{\sqrt{p}} \cdot \frac{\sqrt{\pi}}{2\varepsilon}\big(1 + e^{-(\log p)^2/\varepsilon^2}\big) = -\frac{\log p}{4\sqrt{\pi}\,\varepsilon\sqrt{p}}\big(1 + e^{-(\log p)^2/\varepsilon^2}\big) = \tfrac{1}{2}\,\mathrm{Main}_p(\varepsilon)\big(1 + e^{-(\log p)^2/\varepsilon^2}\big),$$

where the last equality uses $\mathrm{Main}_p(\varepsilon) = -(\log p)/(2\sqrt{\pi}\,\varepsilon\sqrt{p})$. □

Lemma 3.6 (Off-diagonal prime-power bound; proved)**.** *For the off-diagonal prime-power contributions there exist constants $C(p) > 0$ and $c_p > 0$, depending only on p, such that for all $\varepsilon \in (0,1)$,*

$$|\mathrm{PRIMS}_{n\neq p}(\varepsilon)| \;\leq\; \frac{C(p)}{\varepsilon}\, e^{-c_p/\varepsilon^2},$$

where $c_p := \min(d_p^2, (2\log 2)^2)/4 > 0$ and $d_p := \min\{|\log(n/p)| : n = q^k,\, n \neq p\} > 0$.

Remark. *The archimedean and constant-term subleading estimate $|\Gamma_p(\varepsilon)| + \frac{1}{2}|\mathrm{CST}(\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)$ is the content of Assumption 3.2; see the numerical verification in the remark following that assumption.*

Proof. Write $a := \log p$. The prime-power sum $\mathrm{PRIMS}_{n\neq p}$ is bounded by $\sum_{n\neq p} \Lambda(n)/\sqrt{n} \cdot |\hat{h}_{p,\varepsilon}(\log n/(2\pi))|$, and the Fourier transform of $h^{\mathrm{even}}_{p,\varepsilon}$ satisfies

$$|\hat{h}_{p,\varepsilon}(\log n/(2\pi))| \leq \frac{\sqrt{\pi}}{2\varepsilon}\left[e^{-(\log n - a)^2/(4\varepsilon^2)} + e^{-(\log n + a)^2/(4\varepsilon^2)}\right].$$

Split the prime-power sum into the near range $\mathcal{N}_p := \{n = q^m : n \neq p,\, |\log n - a| \leq 1\}$ and its complement.

Near range. $\mathcal{N}_p$ is finite; for each $n \in \mathcal{N}_p$, $|\log n - a| \geq d_p > 0$, so its contribution is $O_p(\varepsilon^{-1} e^{-d_p^2/(4\varepsilon^2)})$.

Far range $|\log n - a| > 1$. For $\varepsilon < 1$,

$$e^{-(\log n - a)^2/(4\varepsilon^2)} \leq e^{-1/(8\varepsilon^2)}\, e^{-(\log n - a)^2/8}.$$

The series $\sum_{n=q^m} \Lambda(n)/\sqrt{n} \cdot e^{-(\log n - a)^2/8}$ converges, since the Gaussian factor $e^{-(\log n)^2/8}$ decays faster than every power of n. Hence the far-range contribution is $O_p(\varepsilon^{-1} e^{-1/(8\varepsilon^2)})$.

Reflected peak. The term $e^{-(\log n + a)^2/(4\varepsilon^2)}$ satisfies $\log n + a \geq 2\log 2$ for all $n, p \geq 2$, so the reflected contribution is bounded by $e^{-(2\log 2)^2/(4\varepsilon^2)}$ times the same summable majorant.

Combining the three pieces and decreasing the exponent constant if necessary gives the bound with $c_p := \min(d_p^2, (2\log 2)^2)/4 > 0$. □

For use in the proof of Theorem 3.1, we record the infinite half-sum form of the Guinand–Weil decomposition. Combining the full Guinand–Weil formula (1) with Assumption 3.3 gives, for fixed prime $p \geq 2$,

$$Z^+_{p,\infty}(\varepsilon) \;=\; \tfrac{1}{2}\,\mathrm{POL}(\varepsilon, p) \;-\; \mathrm{PRIMS}_{n=p}(\varepsilon) \;-\; \mathrm{PRIMS}_{n\neq p}(\varepsilon) \;+\; \Gamma_p(\varepsilon) \;+\; \tfrac{1}{2}\,\mathrm{CST}(\varepsilon) \;-\; \tfrac{1}{2}\,E^{\mathrm{off}}_p(\varepsilon).$$

No finite-N truncation term appears; convergence of the positive-ordinate Gaussian sum is justified by Gaussian decay together with the standard Riemann–von Mangoldt zero-counting estimate (see §, *Analytical foundations*).

Strategy of the proof. The argument proceeds in three steps. *Step 1* substitutes the infinite half-sum form into the definition of $\mathrm{Const}^{\infty}_p$, yielding an exact identity with a pole term, a leading $-\frac{1}{2}\mathrm{Main}_p$ contribution, an exponentially small reflected peak, and a residual R'. *Step*

2 bounds the residual $|R'(p,\varepsilon)| = o(|\mathrm{Main}_p|)$ using Lemma 3.6 and Assumptions 3.2, 3.3. *Step 3* computes the limit $\varepsilon \to 0^+$: the pole term is $O(1)$ against $|\mathrm{Main}_p| \sim \varepsilon^{-1}$, the reflected peak vanishes, and the residual is subleading by Step 2. The dominant term is $-\frac{1}{2}\mathrm{Main}_p$, giving $r_p^\infty = \frac{1}{2}$.

Proof of Theorem 3.1. Step 1: Exact identity. We work with the infinite proxy $Z^+_{p,\infty}$ and $\mathrm{Const}_p^\infty(\varepsilon) = Z^+_{p,\infty}(\varepsilon) - \mathrm{Main}_p(\varepsilon) - \Gamma_p(\varepsilon)$. Applying the Guinand–Weil decomposition (1) to $\widetilde{Z}_p^{\mathrm{GW,full}}$ and then using Assumption 3.3 to transfer to the half-sum $Z^+_{p,\infty}$ (up to an $o(|\mathrm{Main}_p|)$ error), as recorded in the preamble above, together with Lemmas 3.4 and 3.5:

$$\mathrm{Const}_p^\infty(\varepsilon) \;=\; \underbrace{e^{\varepsilon^2/4}\cosh\!\left(\tfrac{\log p}{2}\right)}_{\text{pole, }O(1)} \underbrace{-\tfrac{1}{2}\,\mathrm{Main}_p(\varepsilon)}_{\text{leading, }\sim\varepsilon^{-1}} + \underbrace{\tfrac{1}{2}\,\mathrm{Main}_p(\varepsilon)\,e^{-(\log p)^2/\varepsilon^2}}_{\text{reflected peak, }\to 0} + \underbrace{R'(p,\varepsilon)}_{=o(|\mathrm{Main}_p|)}\,,$$

where $R'(p,\varepsilon)$ collects the off-diagonal prime-power term $-\mathrm{PRIMS}_{n\neq p}(\varepsilon)$, the constant-term contribution $\frac{1}{2}\,\mathrm{CST}(\varepsilon)$, and the proxy-transfer error $-\frac{1}{2}\,E_p^{\mathrm{off}}(\varepsilon)$ from Assumption 3.3. (The $\Gamma_p(\varepsilon)$ contribution from the Guinand–Weil decomposition is absorbed into the definition of Const_p^∞ and does not appear in R'.)

Step 2: Bounding the residual. By Lemma 3.6, $|\mathrm{PRIMS}_{n\neq p}| \le C(p)\varepsilon^{-1}e^{-c_p/\varepsilon^2}$; by Assumption 3.2, $\frac{1}{2}|\mathrm{CST}(\varepsilon)| \le |\Gamma_p(\varepsilon)| + \frac{1}{2}|\mathrm{CST}(\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)$; by Assumption 3.3, $|E_p^{\mathrm{off}}(\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)$. Together these give $|R'(p,\varepsilon)| = o(|\mathrm{Main}_p(\varepsilon)|)$ as $\varepsilon \to 0^+$.

Step 3: Taking the limit. As $\varepsilon \to 0^+$: $e^{\varepsilon^2/4} \to 1$, $e^{-(\log p)^2/\varepsilon^2} \to 0$, so

$$\mathrm{Const}_p^\infty(\varepsilon) \;=\; \cosh\!\left(\tfrac{\log p}{2}\right) - \tfrac{1}{2}\,\mathrm{Main}_p(\varepsilon) + o(|\mathrm{Main}_p(\varepsilon)|)\,.$$

Dividing by $|\mathrm{Main}_p(\varepsilon)| = (\log p)/(2\sqrt{\pi}\,\varepsilon\sqrt{p}) \to \infty$:

$$\frac{\mathrm{Const}_p^\infty(\varepsilon)}{\mathrm{Main}_p(\varepsilon)} \;\longrightarrow\; \frac{\cosh(\log p/2)}{\mathrm{Main}_p(\varepsilon)} - \frac{1}{2} \;\longrightarrow\; 0 - \frac{1}{2} \;=\; -\frac{1}{2}.$$

Therefore $|\mathrm{Const}_p^\infty(\varepsilon)|/|\mathrm{Main}_p(\varepsilon)| \to \frac{1}{2}$ as $\varepsilon \to 0^+$, hence $r_p^\infty = \frac{1}{2}$. ☐

Corollary 3.7 (Sign of asymptotic proxy; proved under Assumptions 3.2 and 3.3)**.** *For every prime $p \ge 2$ there exists $\varepsilon_0(p) > 0$ such that $Z^+_{p,\infty}(\varepsilon) < 0$ for all $\varepsilon \in (0, \varepsilon_0(p))$.*

Proof. By Theorem 3.1 and Assumption 3.2,

$$\frac{|\mathrm{Const}_p^\infty(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|} \to \frac{1}{2}, \qquad \frac{|\Gamma_p(\varepsilon)|}{|\mathrm{Main}_p(\varepsilon)|} \to 0 \qquad (\varepsilon \to 0^+).$$

Choose $\varepsilon > 0$ small enough that

$$|\mathrm{Const}_p^\infty(\varepsilon)| \le \tfrac{3}{4}|\mathrm{Main}_p(\varepsilon)|, \qquad |\Gamma_p(\varepsilon)| \le \tfrac{1}{8}|\mathrm{Main}_p(\varepsilon)|.$$

Since $\mathrm{Main}_p(\varepsilon) < 0$, it follows that

$$Z^+_{p,\infty}(\varepsilon) = \mathrm{Main}_p(\varepsilon) + \Gamma_p(\varepsilon) + \mathrm{Const}_p^\infty(\varepsilon) \le -\big(1 - \tfrac{3}{4} - \tfrac{1}{8}\big)|\mathrm{Main}_p(\varepsilon)| < 0.$$

$\square$

4 Three-Term Decomposition of $O'(\frac{1}{2})$

Definition 4.1 (Three-term decomposition of $O'(\frac{1}{2})$)**.** *Define*

$$\begin{aligned} T_{\mathrm{pol}}(\varepsilon) &:= \tfrac{1}{4}\sum_{p\le\kappa}(\log p)\, e^{\varepsilon^2/4}\sinh\Big(\tfrac{\log p}{2}\Big), \\ T_{\mathrm{main}}(\varepsilon) &:= -\sum_{p\le\kappa}(\log p)\,\frac{1-\frac{1}{2}\log p}{4\sqrt{\pi}\,\varepsilon\sqrt{p}}, \\ T_{\mathrm{res}}(\kappa,\varepsilon,N) &:= O'\big(\tfrac{1}{2};\kappa,\varepsilon,N\big) - T_{\mathrm{pol}}(\varepsilon) - T_{\mathrm{main}}(\varepsilon). \end{aligned}$$

The terms $T_{\mathrm{pol}}(\varepsilon)$ and $T_{\mathrm{main}}(\varepsilon)$ are algebraically explicit in ε and arise from the Guinand–Weil pole and prime main-term contributions, respectively (via Lemma 3.4 and the derivative $\frac{\mathrm{d}}{\mathrm{d}\log p}[(\log p)/\sqrt{p}] = (1-\frac{1}{2}\log p)/\sqrt{p}$). The residual T_{res} collects all remaining terms.

Observation 4.2 (Numerical values; numerical)**.** *At $(\kappa,\varepsilon,N) = (53, 0.05, 100)$: $T_{\mathrm{pol}}(\varepsilon) = +27.67$, $T_{\mathrm{main}}(\varepsilon) = +11.18$, $T_{\mathrm{res}}(53, 0.05, 100) = -36.37$, with sum $O'(\frac{1}{2}) = +2.4751$. The trivial bound $|O'(\frac{1}{2})| \le \mathcal{W}_{\varepsilon,N}\cdot P(\kappa) \approx 28.42\times 44.93 \approx 1277$ exceeds $|O'(\frac{1}{2})|$ by a factor of approximately* 516. *The algebraic terms satisfy $T_{\mathrm{pol}} + T_{\mathrm{main}} = +38.84$, while $T_{\mathrm{res}} = -36.37$; their partial cancellation leaves $|O'(\frac{1}{2})| = 2.4751$, which is a factor of* $38.84/2.4751 \approx 15.7$ *below $|T_{\mathrm{pol}} + T_{\mathrm{main}}|$ [numerical].*

Remark (Domain)**.** *Definition 4.1 uses the ordinate proxy $Z^{+}_{p,N}$, not the full Guinand–Weil object $\widetilde{Z}_p^{\mathrm{GW,full}}$; the GW bridge remains open (Open Problem 9.3).*

5 The Cancellation Hypothesis

The terms $T_{\mathrm{pol}}(\varepsilon)$ and $T_{\mathrm{main}}(\varepsilon)$ of Definition 4.1 are explicitly computable and grow with κ (they are sums over all primes $p \le \kappa$). The residual $T_{\mathrm{res}}(\kappa,\varepsilon,N)$ depends on the fine arithmetic structure of the prime-zero interactions through the exponential sum $S_k(\kappa) = \sum_{p\le\kappa}(\log p)\sin(\gamma_k\log p)$. The following hypothesis asserts a logarithmic saving in $S_k(\kappa)$ relative to its trivial bound $P(\kappa)$.

Assumption 5.1 (Cancellation Hypothesis (A^{**}))**.** *There exist absolute constants $C_0 > 0$ and $A \ge 1$ such that for every Riemann ordinate $\gamma \in \{\gamma_k : k \ge 1\}$ and every prime cutoff $\kappa \ge 2$,*

$$\left|\sum_{p\le\kappa}(\log p)\sin(\gamma\log p)\right| \le \frac{C_0\cdot P(\kappa)}{(\log\gamma)^A}.$$

Two perspectives on (A^{**}). The hypothesis $|S_k(\kappa)| \le C\cdot P(\kappa)/(\log\gamma_k)^A$ admits two equivalent readings of the same sum $S_k(\kappa) = \sum_{p\le\kappa}(\log p)\sin(\gamma_k\log p)$:

Primes interfere at a fixed zero. For each γ_k, the contributions $(\log p)\sin(\gamma_k \log p)$ from individual primes have varying signs. (A^{**}) asserts that these contributions cancel beyond the trivial $P(\kappa)$ envelope by a logarithmic margin.

A zero scans the prime structure. Writing $S_k(\kappa) = \mathrm{Im}\sum_{p\le\kappa}(\log p)\,p^{i\gamma_k}$, each prime acquires a phase via the ordinate γ_k. (A^{**}) asserts that no zero frequency creates strong coherent alignment among the prime phases.

Both readings express the same arithmetic content: *no zero frequency aligns the primes coherently.* This is the analytical question that the present paper isolates but does not resolve.

Remark (Form and strength). *The bound uses $P(\kappa)$ (the Chebyshev theta sum), not $\pi(\kappa)$. The P-form is* weaker *as a hypothesis than the π-form: $P(\kappa) \sim \kappa$ while $\pi(\kappa) \sim \kappa/\log\kappa$, so $P(\kappa)/\pi(\kappa) \sim \log\kappa \to \infty$; the bound on $|S_k(\kappa)|$ with the P-normalisation has a larger right-hand side. The P-form is nonetheless the numerically consistent choice: $|S_k(\kappa)|/P(\kappa)$ stays bounded near $\pm 1/\gamma_k$ [numerical], while $|S_k(\kappa)|/\pi(\kappa)$ grows logarithmically [numerical]. Numerically, $|S_k(\kappa)|/P(\kappa)$ oscillates near $\pm 1/\gamma_k$ [numerical], consistent with a logarithmic saving of size $(\log\gamma_k)^{-1}$. Assumption (A^{**}) asks only for a logarithmic saving in the ordinate aspect, relative to the trivial $P(\kappa)$ bound. It should not be read as a square-root-cancellation statement in the cutoff variable κ; the present paper uses only the displayed logarithmic-saving hypothesis. The non-vanishing $\zeta(1+it) \neq 0$ [21, 22] establishes that there are no zeros on the line $\Re(s) = 1$, but does not by itself imply the logarithmic saving in $S_k(\kappa)$ required by Assumption (A^{**}). The possible relation to Vinogradov–Korobov zero-free-region methods [19, 20] remains open; see Open Problem 9.2.*

Remark (Structural necessity). *Five independent approaches to establishing $|O'(\frac{1}{2})| = o(\mathcal{W}_{\varepsilon,N} \cdot P(\kappa))$ without Assumption (A^{**}) were investigated: (i) Gaussian decay of the Guinand–Weil derivative as $\varepsilon \to 0$; (ii) direct analysis of the cancellation mechanism via Definition 4.1; (iii) the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, which induces $\gamma_k \leftrightarrow -\gamma_k$ (reality of O) but does not induce the symmetry $O(\sigma) = O(1-\sigma)$; (iv) three direct unconditional routes via Weyl equidistribution and trigonometric cancellation; (v) the κ-dependence $O'(\frac{1}{2}; 101, 0.05, 100) = -137.6$, showing that smallness at the reference parameters is a numerical coincidence for this particular κ, not a universal structural property. All five routes yielded only negative results or conditional statements. These failures do not constitute a necessity proof, but they show that the present paper has no unconditional route to stationarity; Assumption (A^{**}) isolates the missing prime-exponential-sum input.*

6 The Integrated-Bias Window

Proposition 6.1 (Integrated-bias window; proved under Assumptions 3.2 and 3.3). *For every prime cutoff $\kappa \ge 2$ and every prime $p \le \kappa$, $Z^{+}_{p,\infty}(\varepsilon) < 0$ for all sufficiently small $\varepsilon > 0$. Consequently there exists $\varepsilon_{\mathrm{int}}(\kappa) > 0$ such that*

$$B^{\infty}_{\mathrm{int}}(\kappa,\varepsilon) := \sum_{p\le\kappa}(\log p)^2\, Z^{+}_{p,\infty}(\varepsilon) < 0$$

for all $0 < \varepsilon < \varepsilon_{\text{int}}(\kappa)$.

Proof. Theorem 3.1 gives $r_p^\infty = \frac{1}{2} < 1$, and Corollary 3.7 gives $Z_{p,\infty}^+(\varepsilon) < 0$ for all $\varepsilon < \varepsilon_0(p)$. Since there are finitely many primes $p \le \kappa$, setting $\varepsilon_{\text{int}}(\kappa) := \min_{p\le\kappa} \varepsilon_0(p) > 0$ gives $Z_{p,\infty}^+(\varepsilon) < 0$ for all $p \le \kappa$ and all $\varepsilon \in (0, \varepsilon_{\text{int}}(\kappa))$. Each summand $(\log p)^2\, Z_{p,\infty}^+(\varepsilon) < 0$, so the sum $B_{\text{int}}^\infty(\kappa,\varepsilon) < 0$. □

Remark. *The curvature identity* $O''(\frac{1}{2}) = -2B$ *involves the* γ_k^2*-weighted sum*

$$B \;:=\; \sum_p (\log p)^2\, \mathrm{Re}\big(Z_p^{(2)}\big), \qquad Z_p^{(2)} \;:=\; \sum_k \gamma_k^2\, e^{-\varepsilon^2\gamma_k^2}\, p^{i\gamma_k},$$

which carries an additional spectral weight absent from B_{int}^∞. *Whether* $B_{\text{int}}^\infty < 0$ *implies* $B < 0$ *is the* Weighted Bias Bridge *(Paper 5, § 6; Open Problem 9.1). At reference parameters* $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $O''(\frac{1}{2}) = +38\,685 > 0$ *is verified numerically.*

Remark (Numerical values)**.** *At the reference parameters* $\varepsilon_{\text{int}}(53) > 0.200$*; numerical computation gives* $\varepsilon_{\text{int}}(2003) \approx 0.086$ *and* $\varepsilon_{\text{int}}(5003) \approx 0.069$ *[numerical].*

Remark (Non-monotonicity)**.** *The function* $\kappa \mapsto \varepsilon_{\text{int}}(\kappa)$ *is non-monotone: sign failures of* B_{int}^∞ *occur at isolated values (*$\kappa = 2003,\ 5003,\ 20011$*) while* $\varepsilon_{\text{int}}(10007) > 0.200$ *[numerical]. This non-monotonicity is driven by a last-prime oscillation effect: at large* ε*, where only* $N_{\text{eff}} = 5$–15 *zeros contribute significantly, the sign of* $B_{\text{int}}^\infty(\kappa,\varepsilon)$ *depends on the Gaussian phase of the final primes before the cutoff. The existence of* $\varepsilon_{\text{int}}(\kappa) > 0$ *is conditional on Assumption 3.2, as Proposition 6.1 establishes.*

7 Conditional Selection Criterion

Proposition 7.1 (Derivative bound; conditional under (A^{**}))**.** *Assume Assumption* (A^{**}) *with constants* $C_0 > 0$ *and* $A \ge 1$*. For all* $\kappa \ge 2$, $\varepsilon > 0$, *and* $N \ge 1$,

$$\big|O'\big(\tfrac{1}{2};\kappa,\varepsilon,N\big)\big| \;\le\; C\cdot \mathcal{W}_{\varepsilon,N}\cdot P(\kappa), \qquad C := \frac{C_0}{(\log\gamma_1)^A},$$

where $\log\gamma_1 \approx 2.6486$ *is independent of* κ,ε,N*. If additionally* $C_0/(\log\gamma_1)^A < 1$*, this bound is genuinely smaller than the trivial bound* $\mathcal{W}_{\varepsilon,N}\cdot P(\kappa)$.

Proof. By Assumption (A^{**}), $|S_k(\kappa)| \le C_0\, P(\kappa)/(\log\gamma_k)^A$ for every $k \ge 1$. Since $O'(\frac{1}{2}) = -\sum_{k=1}^N e^{-\varepsilon^2\gamma_k^2}\gamma_k\, S_k(\kappa)$, we obtain

$$|O'(\tfrac{1}{2})| \;\le\; C_0\, P(\kappa) \sum_{k=1}^N \frac{e^{-\varepsilon^2\gamma_k^2}\,\gamma_k}{(\log\gamma_k)^A} \;\le\; \frac{C_0\, P(\kappa)}{(\log\gamma_1)^A} \sum_{k=1}^N e^{-\varepsilon^2\gamma_k^2}\,\gamma_k \;=\; C\cdot\mathcal{W}_{\varepsilon,N}\cdot P(\kappa),$$

where the second inequality uses $\log\gamma_k \ge \log\gamma_1 > 0$ for all $k \ge 1$. □

Remark. *For fixed $\varepsilon > 0$, the infinite analogue $\sum_{k\geq 1} e^{-\varepsilon^2\gamma_k^2}\gamma_k$ converges by Gaussian decay together with the Riemann–von Mangoldt zero-counting estimate $N(T) \sim T\log T/(2\pi)$ [18, Thm. 5.8]; the bound of Proposition 7.1 therefore also holds in the limit $N \to \infty$.*

Corollary 7.2 (Integrated-bias input; conditional under Assumptions 3.2 and 3.3)**.** *Under Assumptions 3.2 and 3.3, $B_{\mathrm{int}}^{\infty}(\kappa,\varepsilon) < 0$ for $\varepsilon \in (0, \varepsilon_{\mathrm{int}}(\kappa))$ (Proposition 6.1).*

Corollary 7.3 (Second-order local selection criterion)**.** *Fix (κ,ε,N). If $B(\kappa,\varepsilon,N) < 0$ and $O'(\frac{1}{2};\kappa,\varepsilon,N) = 0$, then $\sigma = \frac{1}{2}$ is a strict local minimum of $\sigma \mapsto O(\sigma;\kappa,\varepsilon,N)$, and $\mathrm{Tr}(\widetilde{T}(\sigma))$ has a strict local maximum at $\sigma = \frac{1}{2}$.*

Proof. Under the stated hypotheses, $O'(\frac{1}{2}) = 0$ and the curvature identity $O''(\frac{1}{2}) = -2B$ of [1] gives $O''(\frac{1}{2}) > 0$. The second-order condition then gives a strict local minimum, hence a strict local maximum of $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$. □

The conditional inputs of Proposition 7.1 (Assumption (A^{**})) and Corollary 7.2 (Assumptions 3.2 and 3.3) do not directly imply $B < 0$ or $O'(\frac{1}{2}) = 0$: supplying $B < 0$ requires the Weighted Bias Bridge (Open Problem 9.1), and exact stationarity requires a separate analytical argument. The conditional hierarchy of § 7 therefore stands as follows:

$$\text{Assumptions 3.2 and 3.3} \implies B_{\mathrm{int}}^{\infty} < 0 \qquad \text{(Cor. 7.2);}$$

$$(A^{**}) \implies |O'(\tfrac{1}{2})| \leq C \cdot \mathcal{W}_{\varepsilon,N} \cdot P(\kappa) \qquad \text{(Prop. 7.1);}$$

$$B < 0 \text{ and } O'(\tfrac{1}{2}) = 0 \implies \text{strict local minimum at } \sigma = \tfrac{1}{2} \qquad \text{(Cor. 7.3).}$$

At the reference parameters $(\kappa,\varepsilon,N) = (53, 0.05, 100)$, $B = -19\,342.5 < 0$ is verified numerically; for general parameters, $B < 0$ is the content of Open Problem 9.1.

8 Structural Remarks

8.1 Antisymmetry of the spectral response

Proposition 8.1 (Antisymmetry; proved)**.** *Define $\mathcal{A}(\sigma) := O(\sigma) - O(1-\sigma)$. For all $\sigma \in \mathbb{R}$ and $\delta \in \mathbb{R}$,*

$$\mathcal{A}(\sigma) \;=\; -\sum_{k,p} e^{-\varepsilon^2\gamma_k^2} \sin(\gamma_k \log p)\, \sin\big((2\sigma-1)\gamma_k \log p\big),$$

and in particular $\mathcal{A}(\frac{1}{2}+\delta) = -\mathcal{A}(\frac{1}{2}-\delta)$.

Proof. Apply the difference formula $\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$ to each term of $O(\sigma) - O(1-\sigma)$, with $\alpha = 2\sigma\gamma_k\log p$ and $\beta = 2(1-\sigma)\gamma_k\log p$. One obtains $\alpha - \beta = 2(2\sigma-1)\gamma_k\log p$ and $(\alpha+\beta)/2 = \gamma_k\log p$, yielding the stated formula. The reflection $\sigma \mapsto 1-\sigma$ sends $2\sigma - 1 \mapsto -(2\sigma-1)$, so $\mathcal{A}(1-\sigma) = -\mathcal{A}(\sigma)$; setting $\sigma = \frac{1}{2}+\delta$ gives $\mathcal{A}(\frac{1}{2}+\delta) = -\mathcal{A}(\frac{1}{2}-\delta)$. □

Remark. *The symmetry $O(\sigma) = O(1-\sigma)$ does not hold: one verifies algebraically that $O'(\frac{1}{2}) \neq 0$ in general (numerically, $O'(\frac{1}{2}; 53, 0.05, 100) = +2.4751 \neq 0$). The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ induces a reflection symmetry $\gamma_k \leftrightarrow -\gamma_k$ (reality of O), but not $\sigma \leftrightarrow 1-\sigma$.*

8.2 Remark on unconditional routes to stationarity

Investigation of five independent approaches to establishing $|O'(\frac{1}{2})| = o(\mathcal{W}_{\varepsilon,N} \cdot P(\kappa))$ without Assumption (A^{**}) yielded only negative results or conditional statements (see Remark 5 for details). In particular, the κ-dependence of $O'(\frac{1}{2})$ — which equals -137.6 at $\kappa = 101$ for the same (ε, N) — shows that the smallness at $\kappa = 53$ is a numerical coincidence, not a universal structural identity. The deep cancellation witnessed numerically has the character of a trigonometric sum estimate, and the Cancellation Hypothesis (A^{**}) captures precisely the arithmetic content required.

9 Open Problems

Open Problem 9.1 (Weighted Bias Bridge)**.** Given *(Proposition 6.1, under Assumptions 3.2 and 3.3):*

$$B_{\text{int}}^{\infty}(\kappa, \varepsilon) := \sum_p (\log p)^2 \, Z_{p,\infty}^{+}(\varepsilon) \ < \ 0 \qquad \textit{for } 0 < \varepsilon < \varepsilon_{\text{int}}(\kappa).$$

Target*: the direct curvature sum*

$$B(\kappa, \varepsilon, N) := \sum_p (\log p)^2 \operatorname{Re}\big(Z_p^{(2)}\big),$$

which gives $O''(\frac{1}{2}) = -2B$ via [1].

The bridge requires two successive transfers:

Transfer (i): unweighted $\to$ γ_k^2-weighted asymptotic. *Pass from B_{int}^{∞} to*

$$B^{(2),\infty}(\kappa, \varepsilon) := \sum_p (\log p)^2 \operatorname{Re}\left(\sum_{k \geq 1} \gamma_k^2 \, e^{-\varepsilon^2 \gamma_k^2} \, p^{i\gamma_k} \right).$$

This requires controlling the γ_k^2-weighting in the infinite ordinate sum.

Transfer (ii): asymptotic $\to$ finite-N. *Pass from $B^{(2),\infty}(\kappa, \varepsilon)$ to $B(\kappa, \varepsilon, N)$. This requires a finite-N truncation argument.*

Together, Transfers (i) and (ii) would yield $B < 0$, hence $O''(\frac{1}{2}) > 0$ at the reference parameters. This is the Weighted Bias Bridge *of Paper 5, § 6.*

Open Problem 9.2 (Cancellation Hypothesis)**.** *Establish Assumption* (A^{**}) *analytically. In particular: does there exist a treatment via Vinogradov–Korobov exponential sum techniques (see [19, 20] for the state of the art on zero-free regions) which yields the logarithmic saving in*

$|S_k(\kappa)|/P(\kappa)$ *asserted by Assumption* (A^{**})*?*

Open Problem 9.3 (GW bridge and off-critical control)**.** *The full Guinand–Weil zero sum satisfies*

$$\widetilde{Z}_p^{\mathrm{GW,full}}(\varepsilon) = Z_p^{\infty}(\varepsilon) + E_p^{\mathrm{off}}(\varepsilon),$$

where E_p^{off} *collects contributions from zeros off the critical line (Paper 5, § 6). Establish bounds or sign control on* E_p^{off}*, or identify conditions under which* $E_p^{\mathrm{off}}(\varepsilon) = o(|\mathrm{Main}_p(\varepsilon)|)$ *holds without assuming RH. (This is the same scale as Assumption 3.3; under Theorem 3.1 the two scales* $|\mathrm{Main}_p|$ *and* $|Z_p^{\infty}|$ *are comparable.) The present paper proves* $r_p^{\infty} = \frac{1}{2}$ *for* $Z_{p,\infty}^{+}$ *(under Assumptions 3.2 and 3.3); the unconditional transfer to* $\widetilde{Z}_p^{\mathrm{GW,full}}$ *requires controlling* E_p^{off}*.*

Open Problem 9.4 (Scaling of $\varepsilon_{\mathrm{int}}(\kappa)$)**.** *Determine the asymptotic behaviour of* $\varepsilon_{\mathrm{int}}(\kappa)$ *as* $\kappa \to \infty$*. In particular: is* $\inf_\kappa \varepsilon_{\mathrm{int}}(\kappa) > 0$*? Numerical evidence gives* $\varepsilon_{\mathrm{int}}(53) > 0.200$*,* $\varepsilon_{\mathrm{int}}(2003) \approx 0.086$*,* $\varepsilon_{\mathrm{int}}(5003) \approx 0.069$*,* $\varepsilon_{\mathrm{int}}(10007) > 0.200$ *[numerical], suggesting oscillatory behaviour without monotone decrease, but no analytical lower bound is known.*

Open Problem 9.5 (Near-minimum behaviour at the critical line)**.** *For which parameter pairs* (κ, ε, N) *does the continuous minimum of* $O(\sigma; \kappa, \varepsilon, N)$ *lie near* $\sigma = \frac{1}{2}$*, and under what additional stationarity condition can it occur exactly at* $\sigma = \frac{1}{2}$*? At the reference parameters* $(53, 0.05, 100)$ *one has* $O'(\frac{1}{2}) = +2.4751 \neq 0$*, so* $\sigma = \frac{1}{2}$ *is not an exact continuous local or global minimum. The nearby stationary point lies at* $\sigma_* \approx 0.4999$ *[numerical, resolved local scale]. Exact selection of* $\sigma = \frac{1}{2}$ *requires the separate stationarity condition isolated in Corollary 7.3. For* $\kappa = 101$*,* $\varepsilon = 0.05$ *(fixed* $N = 100$*), the corresponding near-minimum behaviour fails [numerical].*

Open Problem 9.6 (Weil-transfer operator and positivity)**.** *Paper 2 [5] introduces* σ*-dependent admissible Gaussian test functions* $g^*_{\sigma,\varepsilon} \in \mathcal{S}_{\mathrm{ad}}$*, but does not provide a per-prime family. For the present problem, define the prime-local antisymmetric Gaussian component*

$$\psi_{p,\varepsilon}(x) := \phi_\varepsilon(x - \log p) - \phi_\varepsilon(x + \log p),$$

where ϕ_ε *is the Gaussian* $\phi_\varepsilon(x) = (2\pi\varepsilon^2)^{-1/2} \exp(-x^2/(2\varepsilon^2))$*. Define the* Weil-transfer matrix $\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon}$ *by*

$$(\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon})_{pq} \;:=\; W\big(\psi_{p,\varepsilon} * \check{\psi}_{q,\varepsilon}\big)\,,$$

where $W(\cdot)$ *denotes the Weil quadratic functional of [6, 13]. The matrix* $\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon}$ *is a finite-dimensional representation of the Weil form on the admissible subspace* $H^{\mathrm{test}}_{\kappa,\varepsilon} := \mathrm{span}\{\psi_{p,\varepsilon} : p \leq \kappa\}$*.*

(i) *Establish whether the bridge identity holds, where* $c^\eta := (\sqrt{V_p(\frac{1}{2})})_{p \leq \kappa}$ *is the vector of spectral weights introduced in Paper 3 [7]:*

$$\big\langle \mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon}\, c^\eta,\, c^\eta \big\rangle \;=\; \alpha_{\kappa,\varepsilon} \cdot O''\big(\tfrac{1}{2}\big) + E_{\mathrm{off}} + E_{\Gamma/\mathrm{pole}}$$

holds for some normalisation constant $\alpha_{\kappa,\varepsilon}$*, with* E_{off} *an off-diagonal remainder and* $E_{\Gamma/\mathrm{pole}}$ *a* Γ*-factor correction. The four open verification steps are: admissibility of the family* $\{\psi_{p,\varepsilon}\}$

in the Weil sense and its compatibility with the Paper-2 normalisation and Fourier convention; derivation of $\alpha_{\kappa,\varepsilon}$ from Fourier conventions; formal identification of the diagonal $(\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon})_{pp} \sim V_p(\frac{1}{2})$; and control of E_{off}.

(ii) *If (i) holds: can the conditional input $B^{\infty}_{\mathrm{int}} < 0$ from Proposition 6.1, together with an independent proof or finite-grid verification of $O''(\frac{1}{2}) > 0$, be transferred to a Weil-functional shadow $\langle \mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon} c^{\eta}, c^{\eta} \rangle \geq 0$ on a growing finite-cutoff subspace, and does $\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon} \succeq 0$ hold in the limit? Positivity of the Weil form on this subspace would constitute a finite-cutoff analogue of the Weil positivity criterion of [6], which is known to be equivalent to the Riemann Hypothesis [6, 13].*

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The ordinates γ_k are used as numerically specified inputs; no property of their distribution requiring the Riemann Hypothesis is used in any proof. The identity $\widetilde{Z}_p^{\mathrm{GW,full}} = Z_p^{\infty}$ (under RH) is carefully separated from the unconditional computations, which use only the ordinate proxy $Z^{+}_{p,N}$.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The distribution of the zeros is not modelled by random matrix theory. The operator $\widetilde{T}$ is not a Hilbert–Pólya operator (Paper 4, Hilbert–Pólya diagnostic: $r_2 \to 0.16$); its matrix entries are explicit deterministic arithmetic expressions. No pair correlation conjecture or related statistical hypothesis is used.

Conditional inputs. Theorem 3.1, Corollary 3.7, and Proposition 6.1 are conditional on Assumptions 3.2 and 3.3. Proposition 7.1 is conditional on Assumption (A^{**}) (Cancellation Hypothesis). Definition 4.1, Observation 4.2, and Proposition 8.1 are unconditional. Neither Assumption 3.2 nor Assumption $(A^{**})$is known to follow from RH, GUE, or a Hilbert–Pólya postulate. Assumption 3.3 is implied by RH and is formally weaker in content: it requires only asymptotic smallness $o(|\mathrm{Main}_p|)$ of the off-critical contribution, not its vanishing. No converse implication is claimed. The non-vanishing $\zeta(1+it) \neq 0$ [21, 22] is an unconditional analytical input, distinct from both assumptions.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (https://zenodo.org/search?q=Ulrich+Tehrani) and GitHub (https://github.com/utehrani/analysislab-nt).

Topics of particular interest include: analytical approaches to Assumption (A^{**}) via exponential sum methods and Vinogradov–Korobov zero-free region techniques; connections between the integrated-bias window $\varepsilon_{\mathrm{int}}(\kappa)$ and zero-free regions for $\zeta(s)$; and any errors or gaps in the present arguments.

References

[1] U. Tehrani, "Positive Curvature of the Spectral Trace at the Critical Line", Zenodo, https://doi.org/10.5281/zenodo.19665790, 2026.

[2] U. Tehrani, "Spectral Trace Formula and Smoothed Zero Sums: A Prime–Zero Duality Framework", Zenodo, https://doi.org/10.5281/zenodo.19508547, 2026.

[3] U. Tehrani, Verification scripts for the prime–zero coupling series, GitHub, https://github.com/utehrani/analysislab-nt.

[4] U. Tehrani, "A Curvature Decomposition of the Explicit Formula for the Riemann Zeta Function", Zenodo, https://doi.org/10.5281/zenodo.19025598, 2026.

[5] U. Tehrani, "From Local Curvature to the Weil Functional: An Explicit Construction", Zenodo, https://doi.org/10.5281/zenodo.19106992, 2026.

[6] J.C. Lagarias, "On a positivity property of the Riemann ξ-function", *Acta Arithmetica* **89** (1999), 217–234.

[7] U. Tehrani, "A Finite-Cutoff Hilbert-Space Model for Prime–Zero Energy Structure", Zenodo, https://doi.org/10.5281/zenodo.19307989, 2026.

[8] U. Tehrani, "A Dual Operator for Prime–Zero Coupling and a Conditional Proof of Energy Asymmetry", Zenodo, https://doi.org/10.5281/zenodo.19364703, 2026.

[9] M.V. Berry and J.P. Keating, "The Riemann Zeros and Eigenvalue Asymptotics", *SIAM Review* **41**.2 (1999), 236–266.

[10] L. de Branges, "The Riemann Hypothesis for Hilbert Spaces of Entire Functions", *Bull. Amer. Math. Soc. (N.S.)* **15**.1 (1986), 1–17.

[11] A. Connes and C. Consani, "Spectral triples and zeta-cycles," *Enseign. Math.* **69** (2023), no. 1–2, 93–148. arXiv:2106.01715.

[12] A. Connes, C. Consani, and H. Moscovici, "Zeta spectral triples," arXiv:2511.22755 (2025).

[13] E. Bombieri, "Remarks on Weil's Quadratic Functional in the Theory of Prime Numbers, I", *Rend. Lincei Mat. Appl.* **11**.3 (2000), 183–233.

[14] J.-F. Burnol, "On Fourier and Zeta(s)", *Forum Math.* **16**.6 (2004), 789–840.

[15] M. Suzuki, "On the Hilbert space derived from the Weil distribution," *Canad. J. Math.*, First View (2025), 1–24. DOI: 10.4153/S0008414X25101739. arXiv:2301.00421.

[16] A.P. Guinand, "A summation formula in the theory of prime numbers", *Proc. London Math. Soc.* (2) **50** (1948), 107–119.

[17] A. Weil, "Sur les 'formules explicites' de la théorie des nombres premiers", *Comm. Sém. Math. Univ. Lund, Suppl.* (1952), 252–265.

[18] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, AMS, Providence, RI, 2004.

[19] M. J. Mossinghoff, T. S. Trudgian, and A. Yang, "Explicit zero-free regions for the Riemann zeta-function," *Res. Number Theory* **10** (2024), art. 11. DOI: 10.1007/s40993-023-00498-y.

[20] C. Bellotti, "Explicit bounds for the Riemann zeta function and a new zero-free region," *J. Math. Anal. Appl.* **536** (2024), no. 2, art. 128249. DOI: 10.1016/j.jmaa.2024.128249. arXiv:2306.10680.

[21] J. Hadamard, "Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques", *Bull. Soc. Math. France* **24** (1896), 199–220.

[22] C.-J. de la Vallée Poussin, "Recherches analytiques sur la théorie des nombres premiers", *Ann. Soc. Sci. Bruxelles* **20** (1896), 183–256.